

## Geometry 5.5 — Homework Answers

Odd-numbered exercises. Check your work against these; bring questions on the evens to class.

---

1. Corresponding Parts of Congruent Triangles are Congruent
3. any three of:  $\overline{JK} \cong \overline{PQ}$ ,  $\overline{KL} \cong \overline{QR}$ ,  $\overline{LJ} \cong \overline{RP}$ ,  $\angle J \cong \angle P$ ,  $\angle K \cong \angle Q$ ,  $\angle L \cong \angle R$
5. ASA ( $\angle QPR \cong \angle SPR$ ,  $\overline{PR}$  shared,  $\angle QRP \cong \angle SRP$ ) proves  $\triangle QPR \cong \triangle SPR$ ; then  $\overline{PQ} \cong \overline{PS}$  by CPCTC.
7.  $\angle J \cong \angle M$  (alternate interior angles) and  $\angle JLK \cong \angle MLN$  (vertical angles), with  $\overline{JL} \cong \overline{LM}$  between them, so  $\triangle JKL \cong \triangle MNL$  by ASA. Then  $\overline{JK} \cong \overline{MN}$  by CPCTC and  $JK = 34$  ft.